

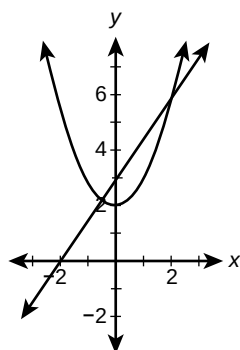
Linear/Quadratic Systems

Linear and quadratic equations can share points that are solutions to both equations.

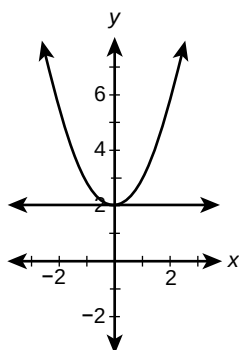
Definition

A **system of equations** is a group of two or more equations in the same variables.

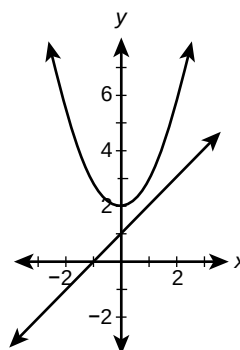
A system of equations that consists of one linear and one quadratic equation can share two points, one point, or no points. The points of intersection for the linear and quadratic graphs represent ordered pairs that are solutions to both equations.



System with
Two Solutions



System with
One Solution



System with
No Solution

The solutions to a linear/quadratic system of equations are ordered pairs. The ordered pairs can be determined by graphing the equations on the coordinate plane or by finding the solution using substitution.

Solving by Graphing

EXAMPLE 1

Use a graph to find the solution to the linear/quadratic system.

$$y = x^2 + 1$$
$$y - x = 1$$

SOLUTION

Create a table of ordered pairs for each equation.

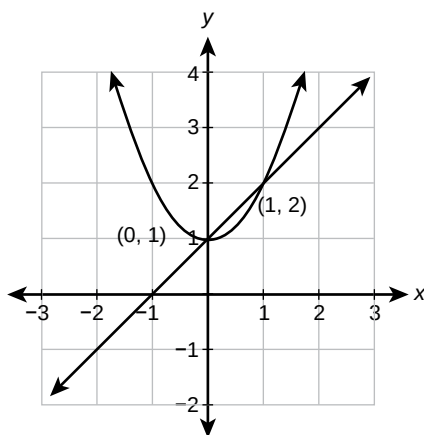
$$y = x^2 + 1$$

x	-2	-1	0	1	2
y	5	2	1	2	5

$$y - x = 1$$

x	-1	0	2	1	4
y	0	1	3	2	2

Graph each equation.



The linear and quadratic equations in the system share two points. The solution to the system consists of the ordered pairs $(0, 1)$ and $(1, 2)$. ■

Solving by Substitution

EXAMPLE 2

Use substitution to find the solution to the linear/quadratic system.

$$\begin{aligned}y &= x^2 - 2x + 2 \\ y - 2x &= -2\end{aligned}$$

SOLUTION

Solve the linear equation for y .

$$y - 2x = -2$$

$$y - 2x + 2x = -2 + 2x \quad \text{Add } 2x \text{ to both sides.}$$

$$y = 2x - 2 \quad \text{Simplify.}$$

Substitute the expression for y into the quadratic equation.

$$y = x^2 - 2x + 2$$

$$2x - 2 = x^2 - 2x + 2 \quad \text{Substitute.}$$

$$2x - 2 - 2x = x^2 - 2x + 2 - 2x \quad \text{Subtract } 2x \text{ from both sides.}$$

$$-2 = x^2 - 4x + 2 \quad \text{Simplify.}$$

$$-2 + 2 = x^2 - 4x + 2 + 2 \quad \text{Add } 2 \text{ to both sides.}$$

$$0 = x^2 - 4x + 4 \quad \text{Simplify.}$$

$$0 = (x - 2)(x - 2) \quad \text{Perfect square trinomial}$$

$$x - 2 = 0 \quad \text{Zero Product Property}$$

$$x - 2 + 2 = 0 + 2 \quad \text{Add } 2 \text{ to both sides.}$$

$$x = 2 \quad \text{Simplify.}$$

Substitute the value $x = 2$ into the linear equation to find y .

$$y - 2x = -2$$

$$y - 2(\mathbf{2}) = -2 \quad \text{Substitute.}$$

$$y - 4 = -2 \quad \text{Simplify.}$$

$$y - 4 + 4 = -2 + 4 \quad \text{Add 4 to both sides.}$$

$$y = 2 \quad \text{Simplify.}$$

The linear and quadratic equations in the system share one point. The solution to the system is the ordered pair $(2, 2)$. ■

EXAMPLE 3

Use substitution to find the solution to the linear/quadratic system.

$$\begin{aligned} y &= x^2 + 3 \\ y - x &= -2 \end{aligned}$$

SOLUTION

Solve the linear equation for y .

$$y - x = -2$$

$$y - x + x = -2 + x \quad \text{Add } x \text{ to both sides.}$$

$$y = x - 2 \quad \text{Simplify.}$$

Substitute the value for y into the quadratic equation.

$$y = x^2 + 3$$

$$\mathbf{x} - \mathbf{2} = x^2 + 3 \quad \text{Substitute.}$$

$$x - 2 - x = x^2 + 3 - x \quad \text{Subtract } x \text{ from both sides.}$$

$$-2 = x^2 - x + 3 \quad \text{Simplify.}$$

$$-2 + 2 = x^2 - x + 3 + 2 \quad \text{Add 2 to both sides.}$$

$$0 = x^2 - x + 5 \quad \text{Simplify.}$$

Use the quadratic formula.

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$a = 1, b = -1, \text{ and } c = 5$$

$$x = \frac{-(-1) \pm \sqrt{(-1)^2 - 4(1)(5)}}{2(1)}$$

Substitute values for a , b , and c .

$$x = \frac{1 \pm \sqrt{1 - 20}}{2}$$

Simplify.

$$x = \frac{1 \pm \sqrt{-19}}{2}$$

Simplify.

The value under the radical is a negative number, so the value of x is not a real number. The system has no real solution. The two equations do not share any points. ■